

MitchellNet

VLANs and Wifi

This is one of the most important things to get right in a VLAN-segmented WiFi setup, and it's simpler than it sounds once you understand the mechanism.

The short answer: SSIDs, not access points, define the VLAN

Each of your UniFi APs broadcasts multiple SSIDs simultaneously — for example:

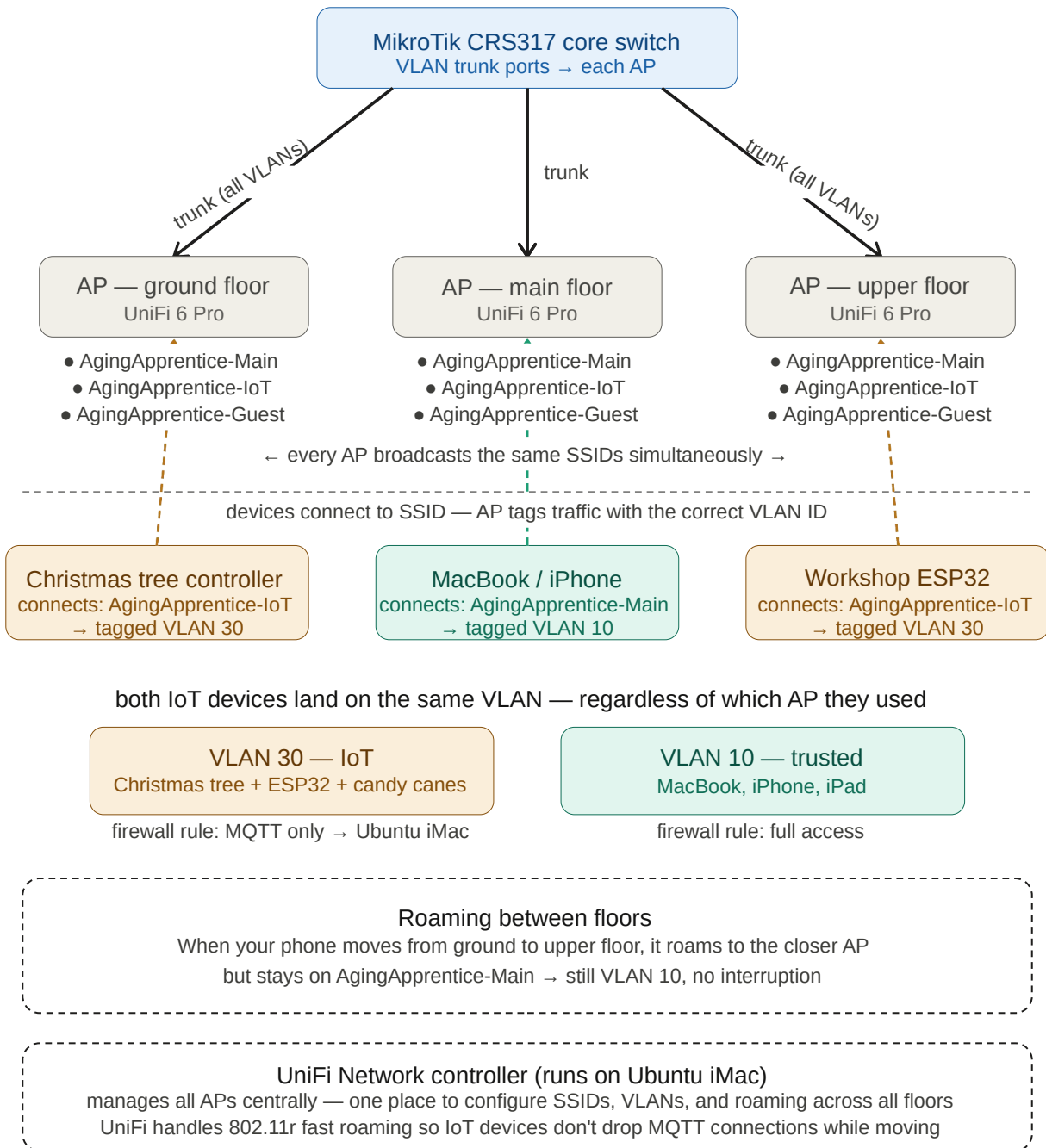
- AgingApprentice-Main → VLAN 10 (trusted devices)
- AgingApprentice-IoT → VLAN 30 (IoT devices)
- AgingApprentice-Guest → VLAN 99 (guest isolation)

Every AP on every floor broadcasts all of those SSIDs at the same time. So when your Christmas tree controller on the main floor connects to `AgingApprentice-IoT`, and your workshop ESP32 on the lower floor also connects to `AgingApprentice-IoT`, they both land on VLAN 30 — regardless of which physical AP they're talking to. The SSID is the ticket, not the AP.

How the tagging actually flows

The key is the cable running from each AP back to your MikroTik core switch. That cable carries a **trunk link** — all VLANs travel down it simultaneously, each tagged with their VLAN ID. When a device connects to the IoT SSID on any AP, the AP tags that traffic with VLAN 30 before sending it up the trunk. The MikroTik sees the tag and routes it accordingly. The firewall then enforces the rules for that VLAN — IoT devices can only reach the Mosquitto broker, nothing else.

Here's a diagram of how it flows:



A few practical points worth knowing:

The UniFi controller runs as a Docker container on your Ubuntu iMac — it's free software called UniFi Network Application. You adopt each AP into it once, and from that single interface you configure every SSID, VLAN tag, and radio setting across all floors simultaneously. You never touch an individual AP again after initial adoption.

For your IoT devices specifically, there's one subtlety: many cheap IoT modules (ESP8266, some ESP32 variants) only support 2.4GHz. Make sure your IoT SSID is broadcast on 2.4GHz on every AP — UniFi lets you pin an SSID to specific radio bands, so you can have the IoT SSID on 2.4GHz only while your main SSID runs on both bands.

Also worth enabling **client isolation** on the IoT SSID. This prevents IoT devices from talking directly to each other over WiFi — they can only communicate through the firewall, which means your candy cane controller can't probe your Christmas tree controller even if they're on the same VLAN. All traffic has to go through OPNsense first.

For the seasonal devices you mentioned — since they're powered off for most of the year, their DHCP leases will expire. When you plug them back in, they'll get a new IP automatically, but because they're on VLAN 30 they'll still land in the right segment and your MQTT topics will still work. The device identity in Grafana/LibreNMS is better tied to MAC address than IP for exactly this reason.